

1. a). $\mathcal{F}[e^{-i\alpha x} f(x)](\alpha) = ?$ + demonstration

b). $f(z) = \frac{1}{z^2 \sin z}$.

Find the singular points and the residues at them.

2. $u = x - 2xy$. Prove that u is harmonic and find v , s.t. $f = u + iv$ is holomorphic, $f(i) = 0$.

3. O equation in general:

$$X(t) + \int_0^t X(s) \cos(t-s) ds = e^t$$

4. $f(t) = \frac{t^2 e^t + \sin^2 2t}{t}$

$\mathcal{L}[f(t)] = ?$